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Data-Driven PHM Solution for Health Monitoring of Mud Motor Power Sections While Drilling

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Abstract

Mud motors are widely used for directional and performance drilling. Due to the extremely challenging operating conditions, they are prone to failures, resulting in unnecessary maintenance repair costs as well as unpredictable and very costly drilling failure.

Until now, the oil and gas industry has lacked reliable procedures to monitor and maintain the health of the mud motor power sections. Recently, we systematically addressed this problem with an industry unique prognostic health management solution, which not only tracks remaining useful life (RUL), but also creates a new failure prevention scheme for operators. The key objective of this solution is to reduce maintenance costs and improve mud motor fleet reliability. It's based on a high-fidelity model and uses a hybrid approach by combining a high-fidelity physics-based model of a power section and data-driven approaches with machine learning techniques for real-time applications.

The new methodology was tested in the field with great success. The verification of the created solution was completed based on numerous field data from Saudi Arabia and Argentina. Comparison of the predicted mud motor fatigue values with the actual observed post-job conditions and job failures demonstrated high fidelity of the developed models. The whole solution is currently being integrated into a drilling platform including the maintenance system, the well construction planning, and the execution. The first application of the workflow was deployed in the field in Colombia targeting reduction of maintenance cost and failure avoidance. The result was outstanding, with the initial deployment bringing about 27% of projected yearly maintenance savings and 10% of projected yearly failure reduction. It enables using the equipment to the full extent with increased drilling performance without sacrificing reliability. In addition, it optimizes the entire fleet management with reduced cost of logistics and maintenance.

The findings of this paper demonstrate the value of the mud motor PHM solution for the oil and gas industry by providing accurate prognosis of power section health, leading to reduced costs, minimized NPT, and increased operational reliability.

Introduction

Mud motors are widely used for directional and performance drilling, as their downhole power capability is still unmatched. Due to the extremely challenging operating conditions, they are prone to failures, resulting

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in unnecessary maintenance repair costs as well as unpredictable and very costly drilling failure. Until now, the oil and gas industry has lacked reliable procedures to monitor and maintain the health of mud motors. Recently, we have addressed this problem with an industry unique health management digital solution, which not only tracks remaining useful life (RUL), but also enables an optimal usage of mud motors during well construction operations. Timely motor retirement and maintenance will ultimately reduce failures, nonproductive time (NPT) and cost of service delivery (COSD).

Proper predictions of mud motor failure and RUL are essential for the timely mud motor maintenance and drilling optimization. The RUL with low uncertainty prolongs the life cycle of mud motors by avoiding redundant maintenance and failures. It also enables the optimized use of the motor power for better ROP with informed decisions about pushing further or not. Overall, it maximizes the usage of the already deployed fleet and reduces the carbon footprint of our operations.

In this paper, we will explore the prognostic health management (PHM) of mud motor power sections. We will start with a quick overview of mud motors, then we will explore the modelling approach for simulating power section behavior before talking about the motor engine, which is a particularly important aspect of the solution and the full PHM.

In the last part of the paper, we will expose the validation scheme for all models before concentrating on the field deployment of the PHM strategy.

Mud Motor Overview

Mud motors are part of downhole assembly and are used to transform the hydraulic energy of the drilling fluid (mud) into mechanical energy on the rotation shaft. Such motors are widely used for directional drilling and performance drilling as their downhole power capability is still unmatched. A mud motor consists of multiple subassemblies consisting of a power section, a transmission, a bearing section, a bent housing, a drive shaft, and top subs. The power section (PS) assembly is the trickiest element where the transformation of hydraulic power into mechanical power occurs. It has two main parts—a rotor and a stator. The rotor is a moving part and is habitually made of steel, and the stator is typically a metal tube with rubber bonded inside. A typical design and parts of a power section can be seen in Figure 1.

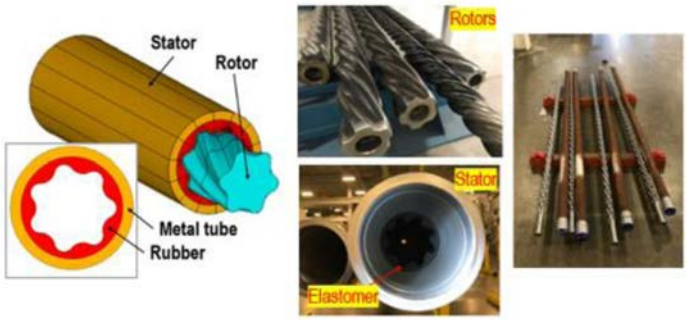

Figure 1—Typical parts of power section assembly

The performance and durability abilities are key features of power sections. Both characteristics can be predicted with good accuracy using numerical models or directly from experiments.

When evaluating the performance of a motor power section, it is in general necessary to refer to power curves. These power curves indicate rpm and torque output of the mud motor depending on the flow rate and differential pressure during operation. Power curves are a very important source of information for drilling engineers or directional drillers since they determine the motor usability. Figure 2 (top) depicts typical power curves for different types of power sections.

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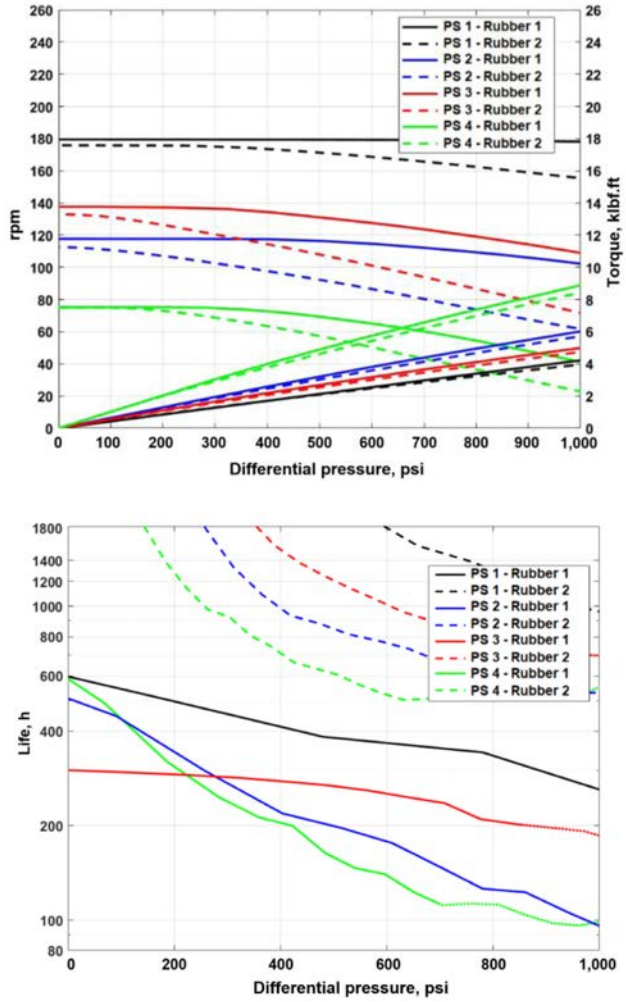

Figure 2—Main characteristics of power sections (PS 1–4): top picture—power curves (rpm/torque vs. differential pressure), bottom picture—fatigue curves (fatigue life of power section vs. differential pressure).

Another important characteristic is the durability. It is often directly linked to fatigue curves. These curves enable understanding of how long the considered power section can be loaded before the elastomer inside the stator starts failing. Fatigue life depends on many parameters and it can be quite different depending on particular power sections and different drilling conditions. Figure 2 (bottom) depicts typical fatigue curves for different types of power sections.

The mud motor failures are most often caused by a failure of the elastomer (rubber) inside the stator, which is exposed to aggressive drilling fluid and must withstand considerable cyclic mechanical loads from the rotor and the pressure of the drilling fluid. Repetitive elastomer deformation is responsible for the growth of fatigue cracks and hysteresis heating, which ultimately can lead to elastomer chunking. Such damage to the power sections can be very costly in that it significantly affects the efficiency of drilling and subsequently the cost of the well construction. That is why the right choice of a suitable power section for a specific drilling case with certain drilling conditions and environment is key.

The fatigue characteristics of power sections depend on the design of the power section and drilling conditions. The main factors that can normally influence their fatigue life are the mechanical design, the elastomer type, the type of drilling fluid, and the drilling conditions (mud flow rate, downhole temperature, differential pressure).

The first two factors are related to the properties of the PS and include the mechanical design of the rotor and stator and the elastomer type inside the stator. The other two factors are related to drilling conditions. The first of the remaining two is the type of drilling fluid that can significantly affect fatigue life due to

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the chemical interaction and the compatibility between drilling fluid and the elastomer. There are many examples where the expected fatigue life for the same motor and the same drilling conditions varies by a factor of two to three due to different drilling fluids.

Modeling approach for simulation of power sections

Despite rather primitive design, power sections of mud motors bring a lot of challenges for modeling and simulation. Full-scale models require multi-physics simulations involving viscoelasticity, hyper-elasticity, and fluid hydrodynamics not to mention the chemical interaction with the fluid. Thus, using full-scale high-fidelity mud motor models implies a lot of computation time and power and a significant amount of preprocessing and postprocessing work and may not be appropriate for all engineering applications.

Although we do have such a high-fidelity model, we also worked on a more efficient model, which takes full advantage of the symmetry of the problem with two assumptions:

- The first assumption is based on the periodicity of power section geometry (Figure 3) and the fact that due to the rotor motion inside the stator, its relative orientation changes cyclically for every cross-section in a similar way as it changes within a so called subpitch (Figure 4).
- The second assumption is reducing the drilling fluid dynamics to the cavity network, which is connected through gaps, and there is no mechanical interaction between the fluid and the elastomer.

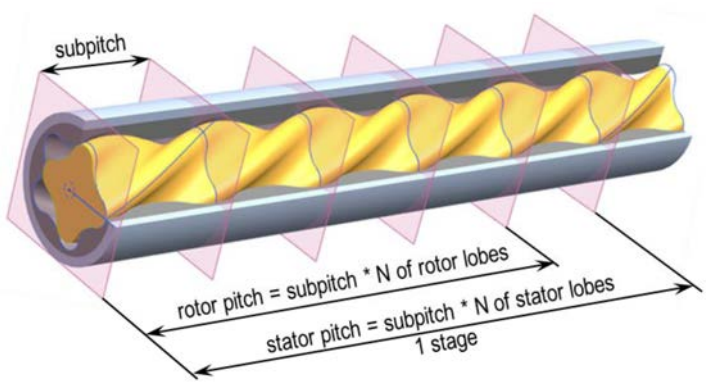

Figure 3—Power section geometry periodicity

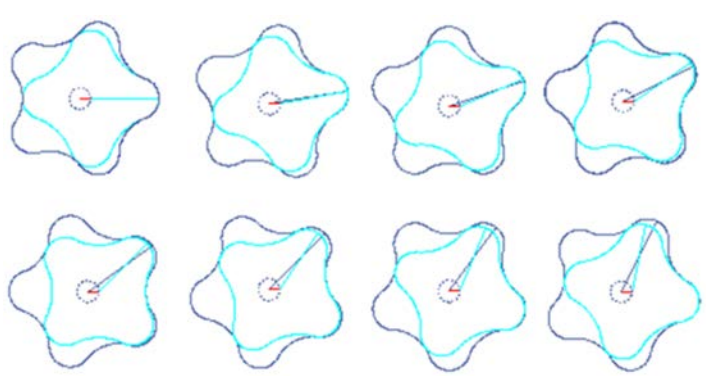

Figure 4—Rotor-stator orientations within one subpitch

So, for the sake of simplicity, a power section is treated as if the displacements, the deformations, and the forces were the same in all cross-sections. The finite element analysis (FEA) is performed for different relative orientation between the rotor and the stator. All the orientations are linked through the rotor orbit constrains. The solutions can be decomposed into time sequences of rotor motion and elastomer

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deformations. This time sequence is looped, i.e., it represents full deformation cycle and is used for hysteresis processing.

At the end, we have a comprehensive modeling tool to predict motor performance and reliability depending on:

- ✓ power section geometry, elastomer, interference fit, downhole temperature, and flow rate.

  The results are output as a numerical table and may include:
- ✓ pressure, torque, rpm, fatigue life in cycle or hours, bonding stress or debonding energy, maximum temperature.

Thus, from an operational point of view, the tool enables selecting an optimal power section, elastomer, and fit for given conditions, which provides both adequate performance and reliability, or performing a cumulative damage analysis to assess the risk of extending the usage of a motor.

Motor Engine

Despite the seemingly faster computation time compared to a full-scale high-fidelity model, a shortage for the simulation tool discussed above is the longer simulation time when compared to the physical time, so its real-time (RT) usage can be challenging. In some scenarios with high RT requirement, a user wants to know the motor power curve and fatigue life every minute or so when the drilling parameters are adjusted while constructing the wellbore.

One possible solution for the RT case is to create a database (look-up table) constructed on precomputed results and use a simple extrapolation to get scenarios which were not included in the database. That way instead of running simulations and waiting minutes or hours to get results, the user would get results in seconds. However, practically, this might be difficult since the number of simulation cases needed in order to obtain acceptable accuracy would be huge due to the nonlinearity of the system, which is not suitable.

A reduced order model is a simplification of a high-fidelity dynamical model that preserves essential behavior and dominant effects with a satisfactory accuracy while reducing computational resources of time and storage. In our case, we used this method and developed a reduced order model for the simulator by using machine learning on the simulated results. A general scheme for the development of the motor engine using machine learning approach is presented in Figure 5; a supervised learning technique is used. Such learning means training the machine learning model using data which is well labeled and have input variables (X) and an output variable (Y). At the end, an algorithm is used to learn the mapping function from X to Y.

$$f: X \rightarrow Y$$

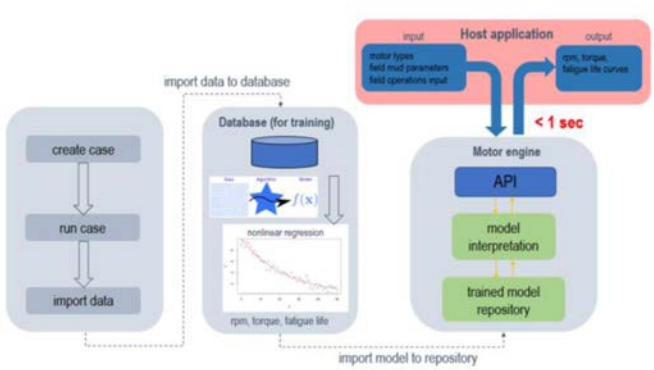

Figure 5—Motor engine scheme

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This very short computation time of the motor engine reduced order model makes it possible now to run multiple simulations for RT applications, which opens the door for RT health monitoring that will be discussed next.

Mud Motor PHM approach

PHM algorithms for the estimation of the RUL of power sections involve considering real field data with significant variations in downhole conditions over time. Drilling conditions (flow rate, differential pressure, and temperature) are key for understanding the workflow of power section health monitoring. Among others, the temperature has the strongest impact on fatigue life. Figure 6 depicts the calculated fatigue life curves for one power section with constant mud flow rate but different downhole operating temperatures. The fatigue life is plotted in logarithmic scale due to the physics of the problem. The results clearly show that the expected operating life for the power section drops dramatically with increasing temperature. If the motor type is known, the fatigue curves for any flow rate and downhole temperature can be easily calculated using the motor engine (III) and then the fatigue life for any differential pressure can be extracted. If changes in the drilling conditions are known at any given time, the expected fatigue life for each set of drilling conditions can be easily predicted from such fatigue curves. The obtained estimations for considered time gaps can be used after that for the RUL calculation by virtue of mathematical integration. Such information about RUL level is an excellent indicator for understanding the power section health.

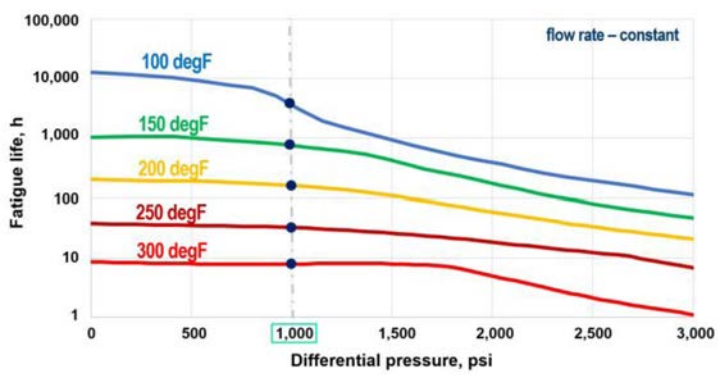

Figure 6—Estimation of PS fatigue life vs. downhole temperature.

In order to compute the health for each time step, it is necessary to take into account the drilling conditions, namely:

- the downhole temperature,
- the flow rate,
- the motor differential pressure,
- the on-bottom status (on/off)
- the bit depth.

The schematic diagram of the workflow for the mud motor PHM can be found in Figure 7. The input parameters are power section characteristics, mud properties, and the drilling conditions mentioned above in the form of RT channels. We can process all this information and calculate the expected fatigue life for each time step. We can track the RUL and consequently get the health of the power section in real time while drilling and after the job for evaluating the rerun criteria or understanding the level of maintenance to be done.

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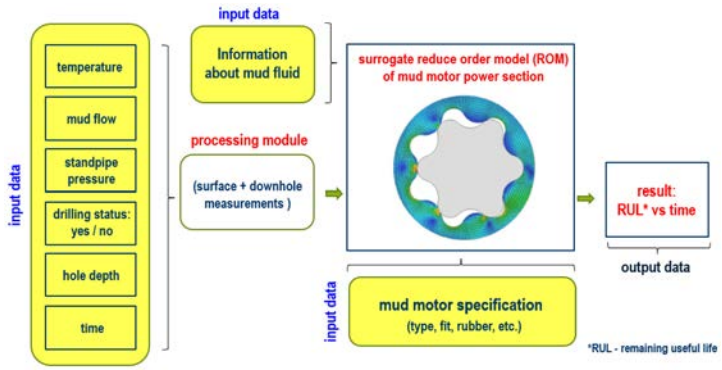

Figure 7—Developed workflow for health monitoring of power sections.

Overall, the whole PHM workflow results in a performance boost as one can adjust the power output on the mud motor in real time and get the maximum possible ROP with informed decisions regarding the remaining useful life of the motor. Obviously, it can also be used to know the typical maintenance strategy as depicted on the last section of this paper.

Validation

Output results of the modelling approach presented in II were validated both in the field and experimentally using a specially designed experimental equipment. This equipment involved taking measurement of over 100 different signals, including pressure at many different locations inside the power section, temperature measurement outside the stator and inside the rubber, torque, flow rate, rpm, lateral displacements at both end of the rotor, shock and vibration at the clamped areas of the stator… More information on the validation process and other details can be found in.

The accuracy of the motor engine was verified for several real field cases in addition to the validation with the original finite element (FE) simulation results described above. Figure 8 depicts the comparative fatigue life curves between the FE simulation and the motor engine: Fatigue life output from the FE approach in red, motor engine fatigue life output in black. The matching between the two is seamless.

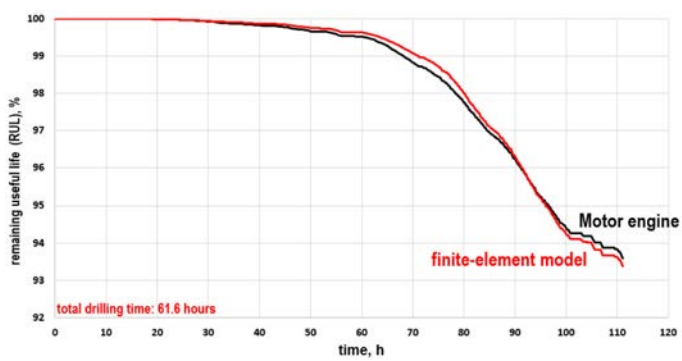

Figure 8—Cumulative remaining useful life curves obtained with help of finite-element simulations (red line) and motor engine (ROM) output (black line).

In addition, the whole mud motor PHM workflow was evaluated on numerous field data at different locations across the globe. Typical results of processed data can be found in Figure 9. The next chapter will discuss the comparison of the predicted fatigue life values with the actual observed post-job conditions and job failures.

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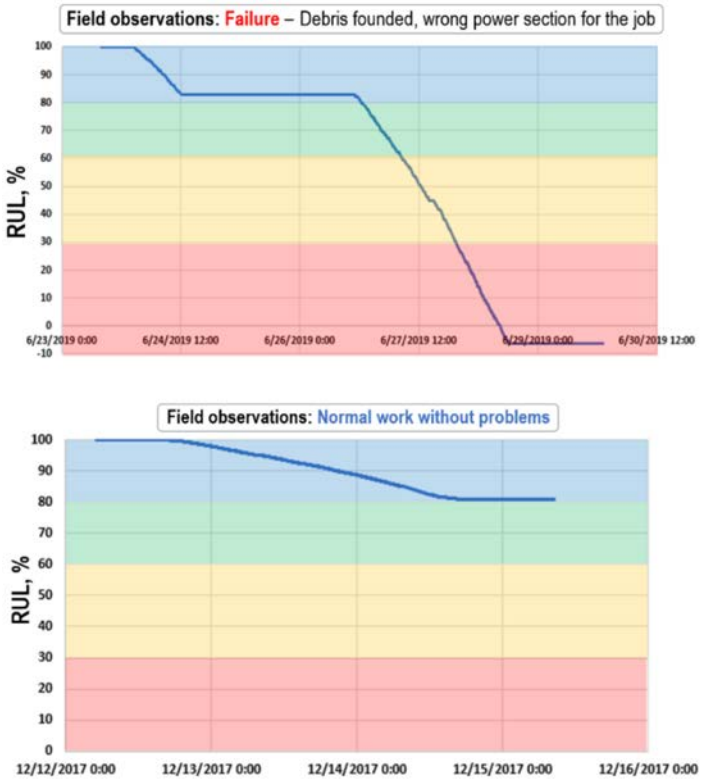

Figure 9—Top picture—example for the case with failure of a power section during the drilling job, bottom picture—Example with successfully used power section. Both figures show the correspondence of the calculated cumulative RUL curves.

Field Implementation of the mud motor PHM

A field implementation of developed mud motor PHM workflow has been started in Q2 2020 and now covers six countries. For the field implementation the locations of mud motor high runners were chosen and segregated according to the following criteria:

- Yearly pumping time (PMPT)
- Coefficients and model availability for the configurations used at the location
- Reason for relining (wear, debris, and chunking)

The locations were requested to follow the decision tree below (Figure 10).

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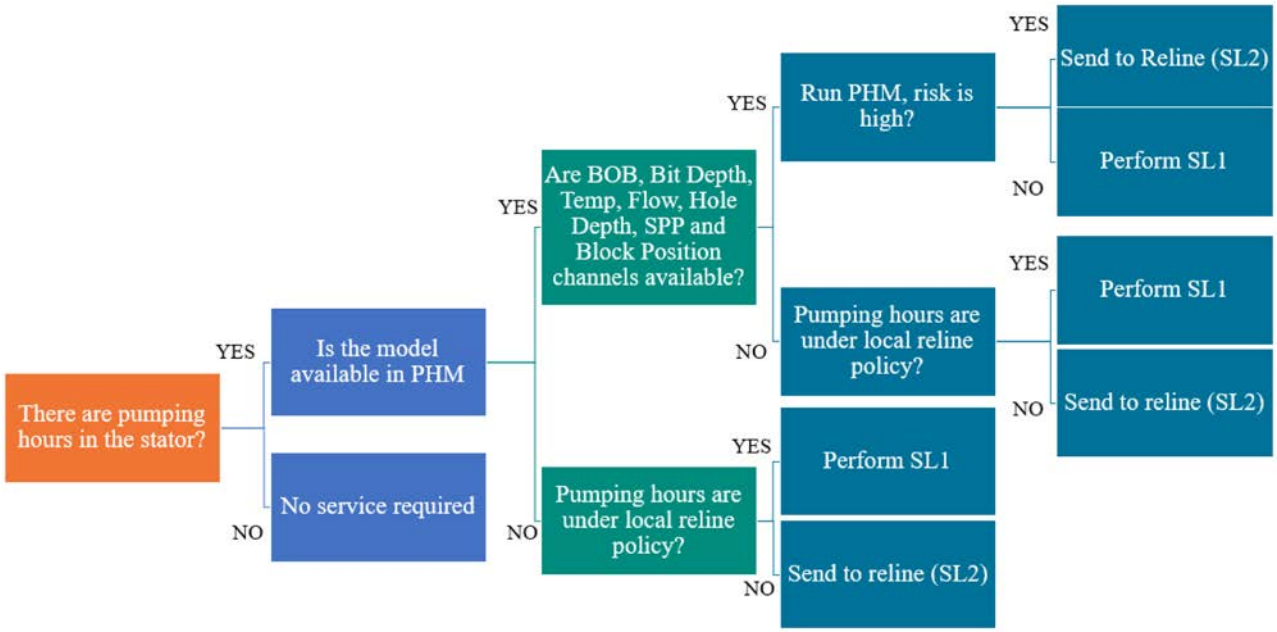

Figure 10—Decision tree for PHM field implementation.

After running the PHM for six months, 202 runs were collected with a total of 8,941.2 hours of pumping time over 74 stators at the different field locations.

An example of the results after running the PHM in one location can be seen in Figure 11.

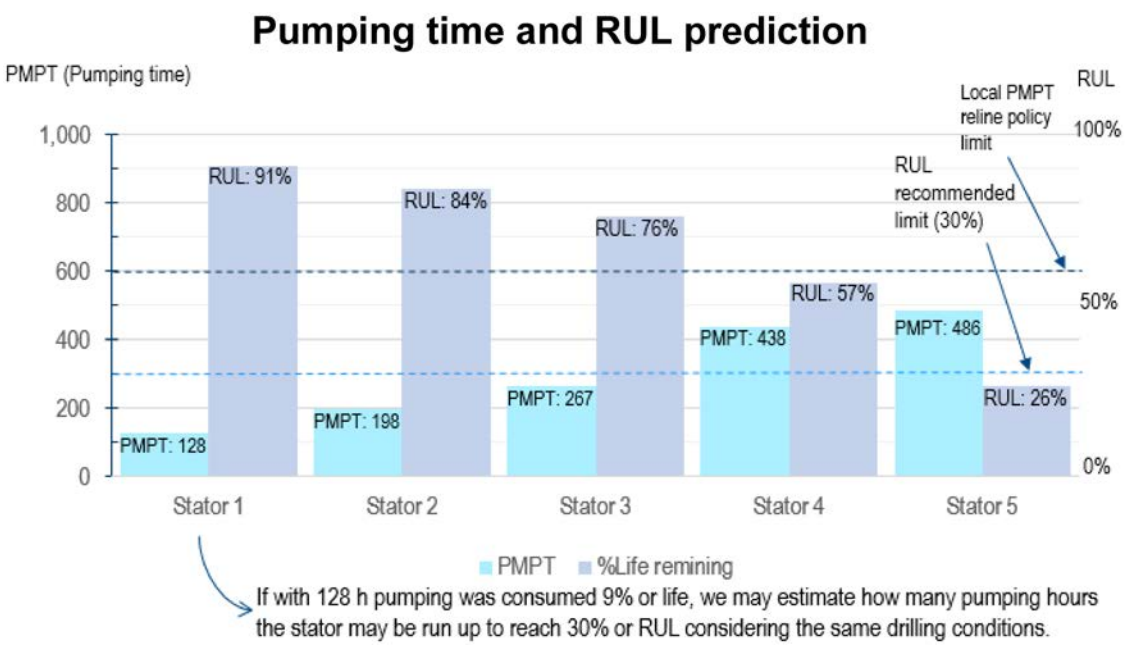

Figure 11—Stators results after running PHM in a location.

We can see that the Stator 4 and 5 in the Figure 11 were run with similar pumping hours, but the life results differ by 31%, showing that the life depends very much on the drilling conditions.

With these initial results, we can extrapolate how long these stators could be run if they are continually running with the same drilling conditions up to the 30% of life:

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Expected duration in the same condition

Figure 12 displays the results for the implementation of the PHM workflow in a particular location. It shows five different stator power sections. The results show that the Stators 1 to 4 have an extended life compared to previously anticipated life (600hrs from the local reline policy). These will result in maintenance savings for the location.

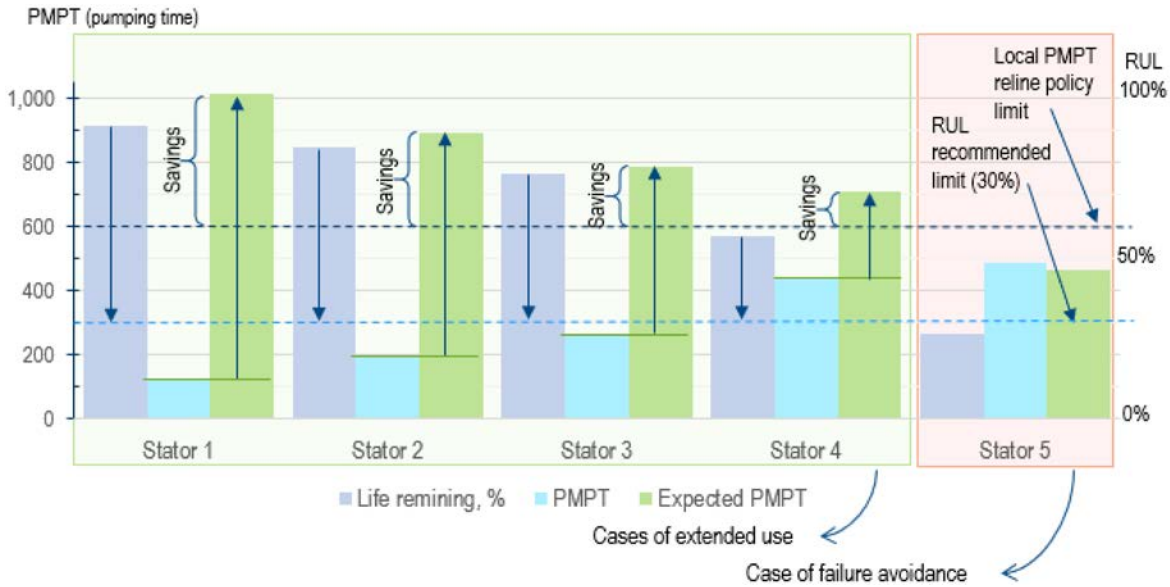

Figure 12—Results showing cases of saving and failure avoidance.

In the other hand, the Stator 5 consumed his life quicker than anticipated. This resulted on savings by avoiding very costly failures. This analysis was done with 43 stators used in that location, showing a projected yearly saving on maintenance (based on reline cost) of 25% and the projected reduction of failure rate by 10%. These are very good results on just this initial deployment for only 43 stators.

Another interesting case is an actual failure during the implementation in another location. Figure 13 displays the calculated RUL by the PHM workflow. It shows that after just one run, the RUL was below the recommended 30%, falling into the red, high risk area.

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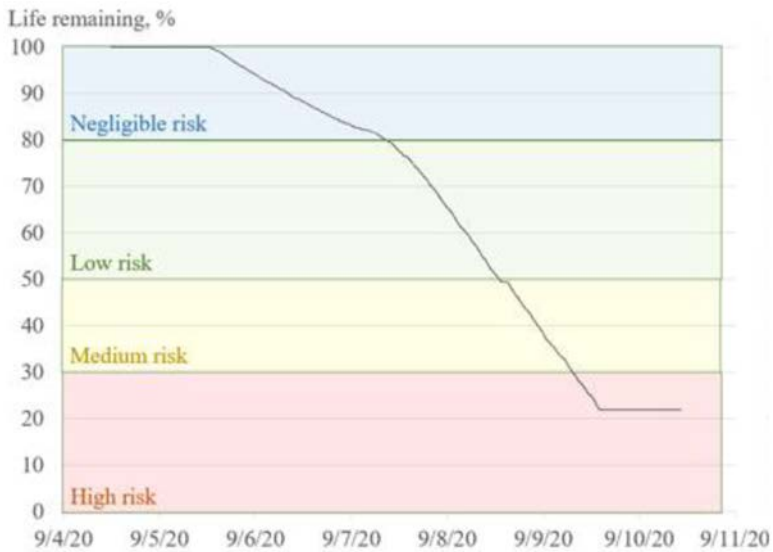

Figure 13—Failed run matching with the PHM prediction

After the run, a picture (Figure 14) of the lower end of the stator shows the beginning of the chunking just like predicted by the PHM workflow even when this was unexpected after just one run

Figure 14—Lower end stator condition, chunking marked.

Conclusion

An accurate model for mud motor PHM has been exposed here. This model uses downhole measurements of temperature plus the surface measurements of differential pressure and flow to estimate the RUL of the elastomer—the life limiting component inside the power section. To enable the use of the model in real-time applications, a reduced order model was derived from an original physics-based model. So, our model uses a hybrid approach that combines a physics-based model and data-driven approaches with machine learning techniques.

The resulted model produces in real time the characterized fatigue and health information in terms of remaining lifespan in percentage along with the output power that can be expected from the motor. This enables ROP optimization without compromising reliability. It can bring value in multiple ways, including the optimization of maintenance operations by minimizing RUL uncertainties, removing motor

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over-maintenance and premature retirements, avoiding costly failure, and more importantly, enabling the adjustment of RT operating parameters for optimized ROP.

The first application of the workflow was deployed in the field targeting reduction of maintenance cost and failure avoidance. The result was outstanding with the initial deployment bringing about 27% of projected yearly maintenance savings and 10% of projected yearly failure reduction.

The next phase on this is to implement it in real-time operations by optimizing the ROP for given applications with informed decision-making process.

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